NUMERICAL METHODS FOR PDES
Lecture 8

Recap from L7:

$$\begin{cases} Lu = -(ku')' + cu = f & (0,1) = \Omega \\ u(0) = 0 = u(1) \end{cases}$$

$k \in L^\infty$, $c \in L^\infty$

(WP) $u \in V := H_0^1(\Omega)$

$$a(u,v) = (f,v) \quad \forall v \in V$$

$$V_h^k = \left\{ u \in C_0(\Omega) : u|_{I_i} \in \mathbb{P}_k(I_i),\ i = 1,\dots,N \right\}$$

$I_i = (x_{i-1}, x_i)$ $\quad x_0 = 0,\ x_i = x_{i-1} + h_i,$

$\dots\ x_N = 1$

$V_h^k \subset V$ "conforming"

for FEM (Galerkin) $u_h \in V_h^k$

If $u \in H^{k+1}(\Omega)$ then

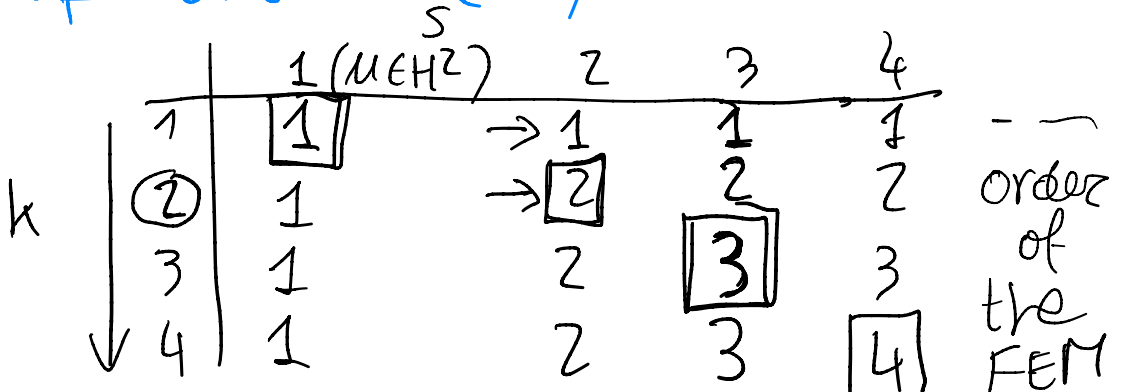
$$= \| (u-u_h)' \|_{L^2} \leq C h^k |u|_{H^{k+1}(\Omega)}$$

$$= |u-u_h|_1 \quad \left(\leq C \left(\sum_i h_i^{2k} |u|^2_{H^{k+1}(I_i)} \right)^{1/2} \right.$$

In fact, if u is less regular, we have reduction in order:

$$|u-u_h|_1 \leq C h^s |u|_{H^{s+1}(\Omega)} \quad \forall\, 1 \leq s \leq k$$

if $u \in H^{s+1}(\Omega)$



Consider $\Omega \subset \mathbb{R}^2$, $\partial\Omega$ smooth



If $f \in H^s(\Omega)$, then

$u \in H^{s+2}(\Omega)$

($\Rightarrow s=0$ i.e $f \in L^2$, then $u \in H^2(\Omega)$)

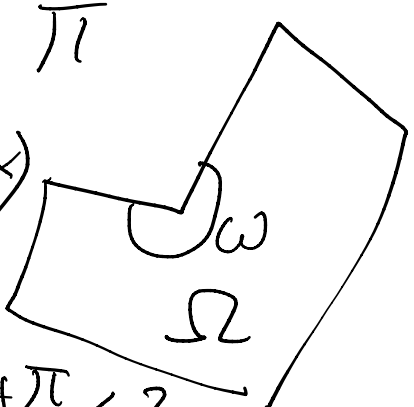
If $\partial\Omega$ is not smooth, say Ω is polygon, then the regularity of u depends on corners.

If ω is the largest corner of Ω and $\omega > \pi$

(i.e. Ω is not convex)

then

$u \in H^s(\Omega) \quad \forall \frac{3}{2} \leq s < 1 + \frac{\pi}{\omega} < 2$



that is $u \in H^2(\Omega)$.

While if Ω is convex, then if $f \in L^2(\Omega) \Rightarrow u \in H^2(\Omega)$.

L^2-error bound

Assume $k \in W^{1,\infty}(\Omega)$.

at least H^2

Theorem: If $u \in H^{s+1}(\Omega)$, for some $s \geq 1$,

$$\|u - u_h\|_0 \leq C h^{k+1} |u|_{H^{k+1}(\Omega)}$$

Proof: (Aubin-Nitsche duality)

Consider the adjoint (dual) problem

$\rightarrow \mathcal{L}^* \phi = e = u - u_h$

(given problem is symmetric, otherwise one should use the adjoint operator, see Q (2.6) and (3.5))

with b.c. $\phi(0) = 0 = \phi(1)$

Weak form: $\phi \in V$ and

adjoint

$$a(w, \phi) = \int_\Omega w\, e \, dx \quad \forall w \in V$$

by hypothesis $\phi \in H^2$ and by

a priori stability,

$$\rightarrow \|\phi\|_{H^2} \leq C \|e\|_0 \qquad \text{(stab)}$$

Let's estimate

$$\|e\|_0^2 = \int_\Omega e \cdot e = a(e, \phi)$$

adjoint with $w = e$

$\|u - u_h\|$

$$= a(e, \phi - I_h \phi)$$

Gal orth.

$$\leq \gamma \|e\|_1 \|\phi - I_h \phi\|_1$$

continuity ↓

$$\leq \gamma \|e\|_1 \left(c h \|\phi\|_2\right)$$

$$\leq C' h \|e\|_1 \|e\|_0$$

(Stab

$$\Rightarrow \|e\|_0 \leq C' h \|e\|_1$$

$$\leq C'' h |h^k| u|_{H^{k+1}(\Omega)}$$

What about $\mathbb{R}^d$ $d>1$

Given $V = H_0^1(\Omega)$, we will consider a partition of Ω.

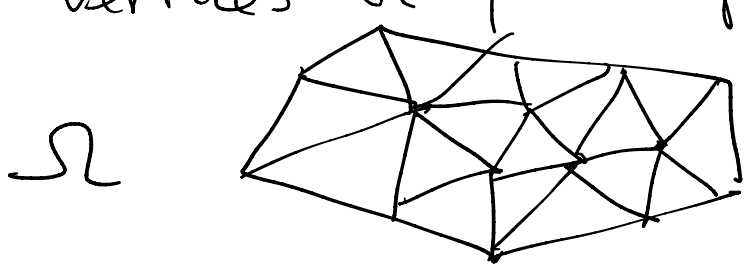
If Ω is a polygon, then

$$\overline{\Omega} = \bigcup T$$

T simplices (triangles if $d=2$)

$\mathcal{T}_h = \{T\}$ assume $T_i \cap T_j$ are

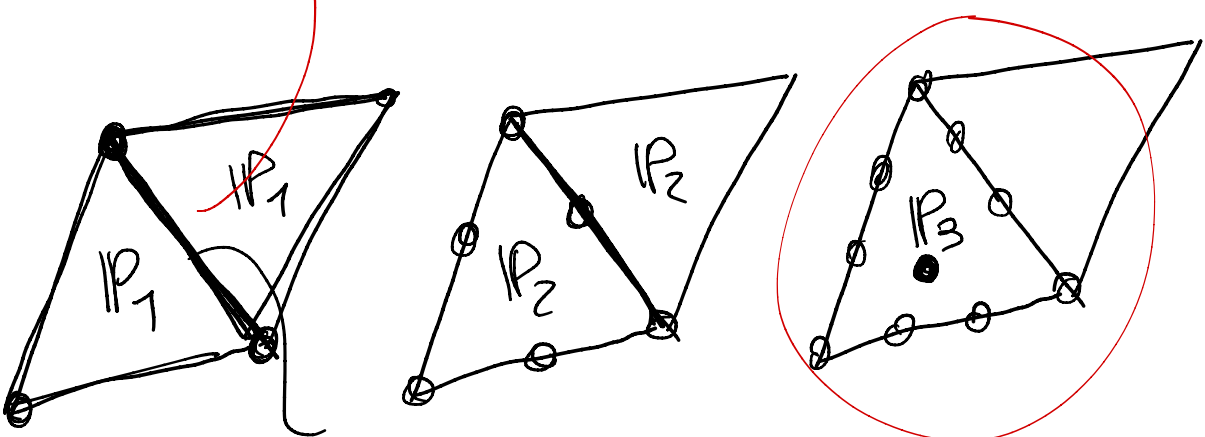
meet at vertices or full edges



(regular partition)

$$V_h^k = \{v \in C_0(\Omega) : v|_T \in \mathbb{P}^k(T) \quad \forall T \in \mathcal{G}_h\}$$

$k=1$ $k=2$ $k=3$



still linear

$$\dim \mathbb{P}_k(\mathbb{R}^d) = \binom{d+k}{k} = \frac{(d+k)!}{k!\,d!}$$

example: $d=2$

$k=$	1	2	3
$\dim \mathbb{P}_k(\mathbb{R}^2)$	3	6	10